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M 435 STEAM / CONDENSATE AND MISCELLANEOUS PIPING SYSTEMS

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M 435 STEAM / CONDENSATE AND MISCELLANEOUS PIPING SYSTEMS

1. SCOPE

The Bidder shall supply the following in accordance with **Part 4, Subsection 1.2 "Summary of Work"**, and this specification

The Bidder shall supply one (1) Steam / Condensate and Miscellaneous Piping Systems for each block (Block 2, 3 & 4).

This specification includes all pipe work between the components in the steam /condensate feed water systems of the Combined Cycle Power Plant specified elsewhere in this specification, as well as the pipe work for other systems not included in any other specification. The scope shall include, but not be limited to, design, supply, transport to site, erection and commissioning of all necessary piping, isolating and control valves, fittings, supports, insulation, instrumentation and other pipe work related accessories.

Pipe work already included in other specifications like **Part 4 Section 3:**

- **M-302 Turbine By-pass System**
- **M-304 Auxiliary Steam System**
- **M-400 Desalination Plant**
- **M-401 Demineralized Water System**
- **M-402 Make up Water System**
- **M-403 Raw Water system**
- **etc.**

shall not be considered under this specification.

This applies for the possible heat cycle alternatives like dual pressure, triple pressure with reheat.

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This specification includes for the pipe work of the systems mentioned below:

- a. HP, LP steam systems as far as applicable
- b. feedwater systems
- c. condensate systems
- d. sealing water system
- e. venting, draining and warming-up systems
- f. Systems which specifications do not include all necessary pipework. These systems shall be listed by the Bidder in his bid. The Bidder shall specify in the list which parts of the pipe work of the specification concerned are included in the price for this specification.

2. CODES AND STANDARD

The Bidder shall comply with the requirements of Part 4, Subsection 2.5 "Codes and Standards" and as mentioned in the specifications referred to.

3. OPERATING AND DESIGN CONDITIONS

Each system shall be designed and installed to provide the appropriate function in accordance with proper engineering and the state of art.

All pumps shall be provided with inlet and outlet isolating valves and minimum flow devices to the approval of the Owner. Outlet valves shall be motorized.

All necessary isolating valves necessary to allow easy maintenance during operation of the plant shall be provided.

Isolation valves shall not be used for control purposes.

Control valves required for continuous plant operation shall be provided with 100% bypass control valves. Both valves, main and bypass control valve, shall be suitably isolated to allow repair during operation of the plant without load reduction and to avoid leakage of the bypass control valve if not in use. Main bypass control valves shall be able to operate automatically, other bypass valves can be hand operated.

4. DETAILED REQUIREMENTS

All piping, valves, supports, etc. shall be in accordance with the following specifications of **Part 4 Section 3**:

- **M-001 Piping**
- **M-002 Pipe Supports and Hangers**
- **M-003 Thermal Insulation and Lagging**
- **M-004 Valves**
- **M-005 Control Valves**

Bidders Data shall be entered in the corresponding Data Sheets in **Part 5 Schedule 7**.

5. PERFORMANCE TESTING

5.1 Performance test

Bidder shall run a performance tests after the equipment is put in service to verify satisfactory performance as required by the Owner.

5.2 Shop test

Refer to **Part 4, Section 3, M-014** and the specifications mentioned herein.

6. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Quality assurance shall conform to the requirements of the above mentioned specifications and of **Part 4, Subsection 2.7**.

7. INSTRUMENTATIONS AND CONTROLS

The Bidder shall be responsible for designing, supplying and installing instrumentation and control devices required for satisfactory operation of the systems and as specified for the connected components.

Pressure and temperature indicators shall be provided at the inlet and outlet of all major components.

Flow measurements shall be provided for each steam flow at the different boiler pressure levels and for the related boiler feedwater flows.

The Bidder shall also be responsible for the complete design, supply and installing of all local instrumentation and control systems, devices, test instruments.

All major flows, pressures and temperatures necessary for suitable operation from the central control room shall be monitored and alarmed in the main control room.

Pipework and system design shall meet the requirements specified in the specification of the connected components, as well as the requirements of the following specifications of **Part 4 Section 3**:

- **I-001 Instrumentation and Control**
- **I-002 Local Control Panels**
- **I-003 Local Mounted Instrument Panels**

8. ELECTRICAL SPECIFICATIONS

8.1 General

For general electrical requirements, see **Part 4 Section 3, E-001**.

8.2 Motors

For detailed electrical requirements, see **Part 4 Section 3, E-210** for Low Voltage Motors.

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